

Question ID 4e375d1f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 4e375d1f

SAT6 M 2.1

How many meters are equivalent to **2,300** centimeters? (**100 centimeters = 1 meter**)

- A. **0.043**
- B. **23**
- C. **2,400**
- D. **230,000**

Question ID bd9eb2b5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: bd9eb2b5

SAT6 M 2.2

The function f is defined by $f(x) = 8x$. For what value of x does $f(x) = 72$?

- A. 8
- B. 9
- C. 64
- D. 80

Question ID 8784bc84

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 8784bc84

SAT6 M 2.3

What is 20% of 440?

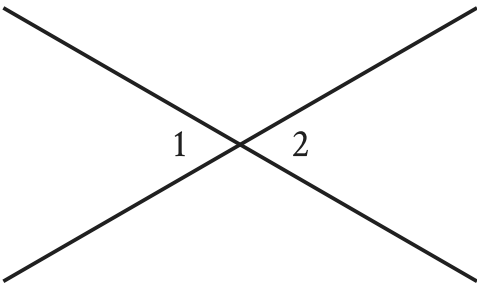
- A. 44
- B. 88
- C. 880
- D. 1,760

Question ID a456f28c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: a456f28c

SAT6 M 2.4



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. Angle **1** and angle **2** are vertical angles. The measure of angle **1** is **72°**. What is the measure of angle **2**?

- A. **72°**
- B. **108°**
- C. **144°**
- D. **288°**

Question ID 51aabd93

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 51aabd93

SAT6 M 2.5

$(p + 3) + 8 = 10$ What value of p is the solution to the given equation?

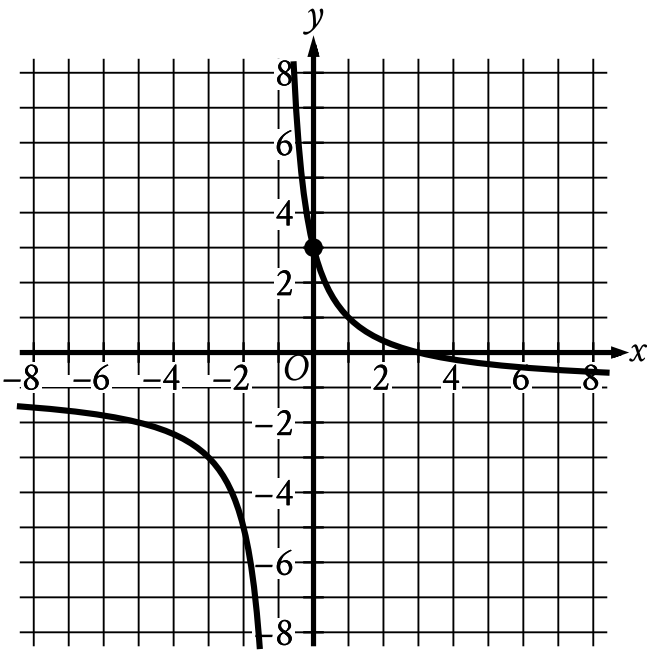
- A. -1
- B. 5
- C. 15
- D. 21

Question ID c99d154a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: c99d154a

SAT6 M 2.6



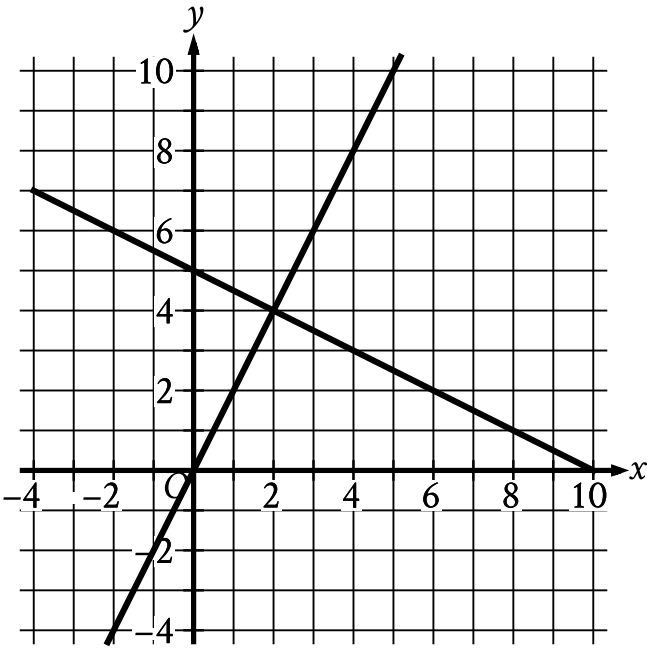
What is the y -coordinate of the y -intercept of the graph shown?

Question ID 1006cad7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 1006cad7

SAT6 M 2.7



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

- A. $(0, 5)$
- B. $(2, 4)$
- C. $(5, 10)$
- D. $(10, 0)$

Question ID 7399c3b0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 7399c3b0

SAT6 M 2.8

$k^2 - 53 = 91$ What is the positive solution to the given equation?

- A. 144
- B. 72
- C. 38
- D. 12

Question ID aacc834b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: aacc834b

SAT6 M 2.9

$x = 8$ $x + 3y = 26$ The solution to the given system of equations is (x, y) . What is the value of y ?

Question ID 842cec4d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 842cec4d

SAT6 M 2.10

During a portion of a flight, a small airplane's cruising speed varied between **150** miles per hour and **170** miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

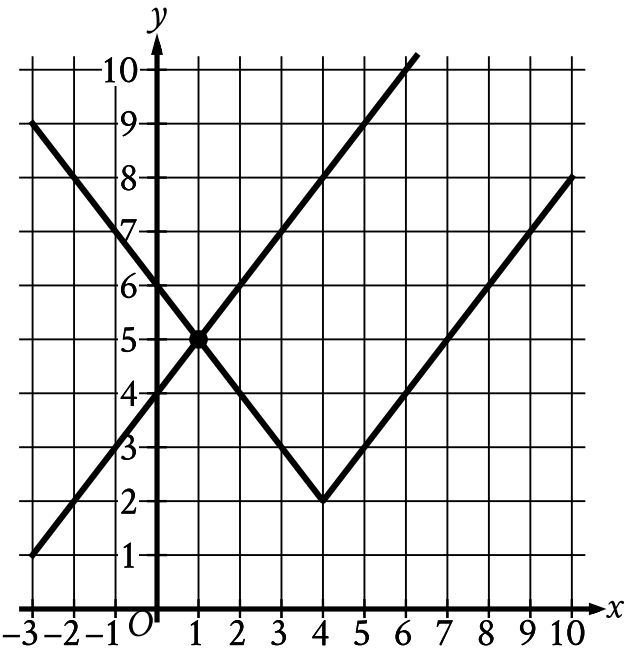
- A. $s \leq 20$
- B. $s \leq 150$
- C. $s \leq 170$
- D. $150 \leq s \leq 170$

Question ID dd3a910a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: dd3a910a

SAT6 M 2.11



The graph of a system of an absolute value function and a linear function is shown. What is the solution (x, y) to this system of two equations?

- A. $(-1, 5)$
- B. $(0, 4)$
- C. $(1, 5)$
- D. $(4, 2)$

Question ID c1eead73

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: c1eead73

SAT6 M 2.12

The function g is defined by $g(x) = |x + 18|$. What is the value of $g(4)$?

- A. -18
- B. -4
- C. 14
- D. 22

Question ID 9fc31513

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 9fc31513

SAT6 M 2.13

The perimeter of an isosceles triangle is **36** feet. Each of the two congruent sides of the triangle has a length of **10** feet. What is the length, in feet, of the third side?

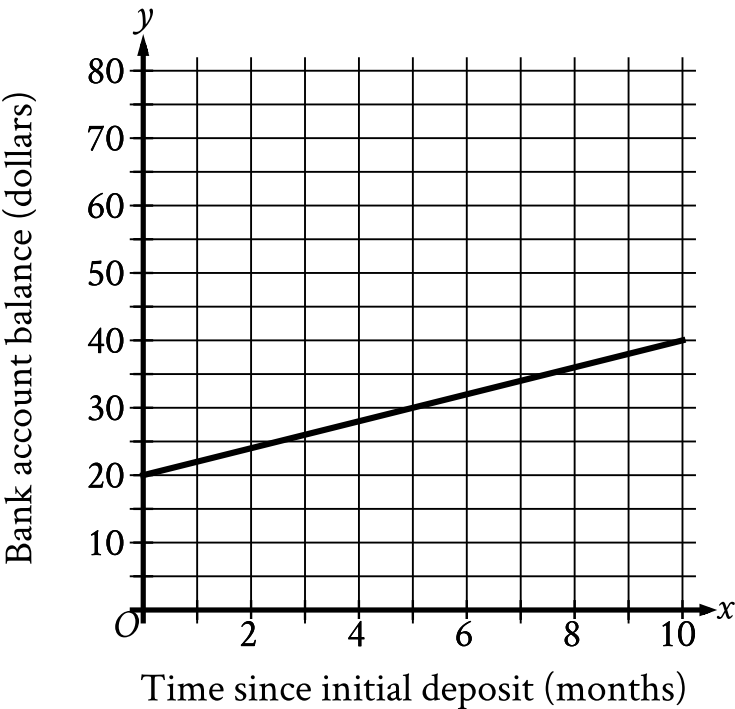
- A. 10
- B. 12
- C. 16
- D. 18

Question ID 2ecce641

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 2ecce641

SAT6 M 2.14



A bank account was opened with an initial deposit. Over the next several months, regular deposits were made into this account, and there were no withdrawals made during this time. The graph of the function f shown, where $y = f(x)$, estimates the account balance, in dollars, in this bank account x months since the initial deposit. To the nearest whole dollar, what is the amount of the initial deposit estimated by the graph?

Question ID 1fc4f47b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 1fc4f47b

SAT6 M 2.15

At a movie theater, there are a total of **350** customers. Each customer is located in either theater A, theater B, or theater C. If one of these customers is selected at random, the probability of selecting a customer who is located in theater A is **0.48**, and the probability of selecting a customer who is located in theater B is **0.24**. How many customers are located in theater C?

- A. **28**
- B. **40**
- C. **84**
- D. **98**

Question ID ebd457f4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: ebd457f4

SAT6 M 2.16

Brian saves $\frac{2}{5}$ of the **\$215** he earns each week from his job. If Brian continues to save at this rate, how much money, in dollars, will Brian save in **9** weeks?

Question ID f28944ff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f28944ff

SAT6 M 2.17

$q(x) = 32(2^x)$ Which table gives three values of x and their corresponding values of $q(x)$ for function q ?

- A.

x	-1	0	1
$q(x)$	-64	0	64
- B.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	2	64
- C.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	32	64
- D.

x	-1	0	1
$q(x)$	16	32	64

Question ID ae33e0a1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: ae33e0a1

SAT6 M 2.18

A right circular cylinder has a base diameter of **22** centimeters and a height of **6** centimeters. What is the volume, in cubic centimeters, of the cylinder?

- A. **132** π
- B. **264** π
- C. **726** π
- D. **2,904** π

Question ID e0f59119

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: e0f59119

SAT6 M 2.19

What is the slope of the graph of $y = \frac{1}{3}(29x + 10) + 5x$ in the xy -plane?

Question ID 44b2b894

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	■ ■ ■

ID: 44b2b894

SAT6 M 2.20

A rectangle is inscribed in a circle, such that each vertex of the rectangle lies on the circumference of the circle. The diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $1,089\sqrt{3}$ square units. What is the length, in units, of the diameter of the circle?

Question ID e723bd67

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: e723bd67

SAT6 M 2.21

$2x - y > 883$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
440	0
441	−2
442	−4

B.

x	y
440	0
442	−2
441	−4

C.

x	y
442	0
440	−2
441	−4

D.

x	y
442	0
441	−2
440	−4

Question ID 9a216c0f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 9a216c0f

SAT6 M 2.22

The combined original price for a mirror and a vase is **\$60**. After a **25%** discount to the mirror and a **45%** discount to the vase are applied, the combined sale price for the two items is **\$39**. Which system of equations gives the original price *m*, in dollars, of the mirror and the original price *v*, in dollars, of the vase?

- A. $m + v = 60$ $0.55m + 0.75v = 39$
- B. $m + v = 60$ $0.45m + 0.25v = 39$
- C. $m + v = 60$ $0.75m + 0.55v = 39$
- D. $m + v = 60$ $0.25m + 0.45v = 39$